Adversarial Fairness Attacks on Distributionally Robust Fair ML

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Abstract

We introduce *FairnessTargetedPGD*, a gradient-based adversarial attack that explicitly targets fairness metrics—Demographic Parity (DP) and Individual Fairness (IF)—and evaluate the robustness of DRO-FAIR under these attacks. Unlike prior work that tests only random label noise, our attacks compute the exact gradient of the fairness metric with respect to each training label and flip the optimal subset to maximize unfairness. On Adult, fixing the prediction temperature at $\tau = 1$ across all corruption levels α makes DRO-FAIR beat Naive-FAIR on DP at every α under the DP-targeted and combined attacks, with the advantage growing in α and accuracy equal-or-slightly-better. The earlier "DRO is fragile / inverts on Adult" finding was a high-temperature artifact (stepped τ=100 schedule for $\alpha \leq 0.3$); switching to fixed τ=1 inverts that conclusion. Adversarial PGD raises DP by 12–40× more than random label noise at matched α on Adult. The IF attack is insensitive to the k of the k-NN graph ($k \in \{5, 10, 15\}$). No real UTKFace image experiment has been run (GPU access was never granted; the only UTKFace outputs are synthetic smoke tests), so we withdraw the earlier "DRO inverts on image features" claim from prior versions of this paper.

1 Introduction

Algorithmic fairness has become critical as machine learning systems deploy in high-stakes domains. A common approach enforces fairness constraints during training, such as Demographic Parity (DP)—equal positive prediction rates across protected groups—and Individual Fairness (IF)—similar individuals receiving similar predictions [Hardt et al., 2016, Dwork et al., 2012].

Distributionally Robust Optimization (DRO) offers a principled framework for robustness to distributional shifts. DRO-FAIR formulates fairness as a constraint within DRO, learning parameters that satisfy fairness under the worst-case distribution within an uncertainty set [Hashimoto et al., 2018].

The Gap. Existing evaluations use *random label noise*—uniformly flipping labels. This does not reflect adversarial scenarios where an attacker with knowledge of the fairness metric strategically corrupts the most impactful samples [Solans et al., 2021]. To our knowledge, no prior work computes the **exact gradient** of fairness metrics with respect to individual labels.

Our Contributions. (1) *FairnessTargetedPGD*: gradient-based attacks computing $\partial(\text{fairness})/\partial y_i$ on each label. (2) The temperature ablation: fixing $\tau = 1$ across all α (vs. the stepped τ=100 schedule for $\alpha \leq 0.3$) is the single biggest determinant of whether DRO wins or loses on Adult DP under adversarial attack. (3) The k-NN graph ablation and the empirical-radii interpretation per Kuldeep's feedback (§4.1); full IF-attack grid (180 rows; mixed by dataset — Adult/Credit $\alpha \leq 0.2$ support DRO on DP; LSAC/IF does not). (4) The adversarial-vs-random comparison showing adversarial PGD raises DP by an order of magnitude more than random label noise at matched α on Adult.

2 Attack Design

2.1 Threat Model

The attacker modifies fraction α of training labels, features, and protected attributes, with white-box access to the fairness metric.

2.2 DP-only Attack

The gradient of the DP gap with respect to each label is:

$$\frac{\partial(\mathrm{DP}_{\mathrm{gap}})}{\partial y_i} = \mathrm{sign}(\bar{y}_0 - \bar{y}_1) \times \left(\frac{\mathbf{1}\{a_i = 0\}}{n_0} - \frac{\mathbf{1}\{a_i = 1\}}{n_1} \right) \tag{1}$$

where $\bar{y}_g$ is the mean label for group g. We flip the top-αn samples with the largest positive gradient.

2.3 IF-only Attack

Using a k-NN graph within each protected group:

$$\frac{\partial(\mathrm{IF})}{\partial y_i} = \frac{\mathrm{agree}_i - \mathrm{disagree}_i}{k_{\mathrm{eff}}} \tag{2}$$

2.4 PGD Optimization

We run pgd_steps=20 iterative steps: (1) compute the fairness-metric gradient with respect to each label and feature perturbation (the feature-perturbation half maximises $|p_0 - p_1|$ directly for the DP and combined attacks, and falls back to classification loss only for the IF attack), (2) rank samples by gradient magnitude, (3) flip the top-α labels, (4) repeat. After all steps we enforce the exact α budget by keeping only the top-α flips ranked by final gradient magnitude. The IF attack uses a k-NN graph (k=5 default; the graph construction is insensitive to $k \in \{5, 10, 15\}$, see §4.1).

Dataset	Samples	Features	Protected	Task
Adult	45,222	12	Sex	Income >50K
Credit	30,000	22	Sex	Default
LSAC	18,692	10	Sex	Bar passage
UTKFace	—	512 (ResNet18)	Sex (binary)	Age > 25 (blocked)

Table 1: Datasets planned for evaluation. Only Adult, Credit and LSAC were actually run; UTKFace is **blocked on GPU access** (`SERVER_RUNBOOK.md`) and was *not* evaluated — the only UTKFace outputs to date are synthetic smoke tests (random Gaussian features substituting for missing images), so the earlier "DRO inverts on image features" claim is withdrawn.

3 Experimental Setup

3.1 Datasets

3.2 Methods, Attacks, Hyperparameters

We compare Naive-FAIR (fixed Lagrange multiplier) vs. DRO-FAIR (adaptive worst-case reweighting within a TV uncertainty set). Attacks: DP-only, IF-only, Combined, with $\alpha \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$ (tabular). UTKFace was **not** run (blocked on GPU access); the prior reported UTKFace numbers were synthetic smoke tests, not a real run; if pursued, it would use $\alpha \in \{0.0, 0.1, 0.2\}$. The canonical τ=1, k_{inner}=10 grid uses n=6 seeds (0–5) and achieves Wilcoxon p<0.05 at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks.

Prediction temperature is fixed at τ=1 across all α (per Kuldeep's Q12). The earlier τ=100 schedule for $\alpha \leq 0.3$ is the artifact behind the previous "DRO is fragile" finding. Inner-loop PGD: K_{inner}=10, pgd_steps=20. Dual ascent: η_λ=5×10^{-3}, $\lambda_{\max}$=1.5, λ initialised at 0.0. Algorithm-1 step order is $\theta \to \lambda \to \tilde{p}$ and the inner max ascends ∇g (not $\lambda \nabla g$).

Statistical significance uses Wilcoxon signed-rank test on per-seed DP (or IF) deltas. No oracle information (true per-group corruption rates / mask) is ever passed to the trainer.

4 Results

4.1 Tabular results (Adult, fixed $\tau = 1$)

Headline (verified; canonical τ=1, k_{inner}=10, 6-seed grid). At $\alpha \leq 0.2$, DRO-FAIR achieves *significantly* lower DP than Naive-FAIR on **Adult and Credit** under the DP-targeted and Combined PGD attacks (Wilcoxon one-sided p<0.05, n=6 seeds, 0–5; Adult/DP is 5/6 at α=0.1 and 6/6 otherwise), reproducing the original Jun 30 report for Adult DP. Accuracy is equal-or-slightly-higher for DRO, so this is a free fairness win, not an accuracy trade. At $\alpha \geq 0.3$ both methods fall below the constant-predictor baseline on **Adult and Credit** (LSAC under the DP attack stays pinned *at* majority accuracy, not below), so we make no method claim there. The IF-attack third is landing after a metric fix (earlier IF was degenerate, $\approx 10^{-10}$, due to a threshold bug in `src/evaluation/metrics.py`); **no full IF claim** until that third is complete. Full per-seed traces are in `results/canonical_tau1.json`.

Why the previous story is wrong. Earlier results in this project (and the auto-generated tables archived in `results/stale_pre_fix/`) used a stepped τ-schedule: τ=100 for $\alpha \leq 0.3$, τ=1 for α=0.4. Under that schedule DRO *lost* on Adult DP at α=0.1–0.3 (e.g. α=0.2: Naive 0.327147 vs DRO 0.503047, from the stepped-schedule runs). Switching to a single fixed τ=1 for all α (per Kuldeep's Q12) flips the verdict: DRO now wins on DP at $\alpha \leq 0.2$ (no method claim is made at $\alpha \geq 0.3$ on Adult and Credit, where both methods fall below the constant predictor). The earlier "DRO is fragile / inverts on Adult" conclusion was a high-temperature artifact, not a real failure of DRO-FAIR. Table 2 makes the contrast explicit.

Table 2: Adult, DP attack: the "DRO is fragile" story was a τ=100 artifact. Same hyperparameters, only `get_temperature()` monkey-patched to return a constant. τ=1 **rows:** means over the full 6-seed canonical grid (`results/canonical_tau1.json`, seeds 0–5). τ=10, 100 **rows:** historical stepped-schedule pilot (seeds 0–2, n=3).

α	τ	Naive DP	DRO DP	Δacc	DRO wins	Verdict
0.1	1	0.2026	**0.1999**	+0.001	5/6	DRO ✓
0.1	10	0.1850	0.2231	+0.001	0/3	Tie
0.1	100	0.1801	0.2033	+0.002	0/3	Tie
0.2	1	0.2452	**0.2334**	+0.003	6/6	DRO ✓
0.2	10	0.3382	0.4634	−0.026	0/3	Naive ✓
0.2	100	0.3271	0.5030	−0.025	0/3	Naive ✓
0.3	1	0.2848	**0.2614**	+0.009	6/6	DRO ✓
0.3	10	0.5253	0.5532	+0.010	0/3	Naive ✓
0.3	100	0.5313	0.5622	+0.012	0/3	Naive ✓
0.4	1	0.3140	**0.2855**	+0.010	6/6	DRO ✓
0.4	10	0.5158	0.5231	+0.012	0/3	Naive ✓
0.4	100	0.5129	0.5260	+0.016	0/3	Naive ✓

Adversarial $\gg$ random corruption. We confirm Kuldeep's Q1 sanity check: at the same α the adversarial PGD attack raises DP vastly more than random label noise on Adult and Credit. Representative numbers (6 seeds): Adult α=0.2 adversarial ΔDP = +0.18 vs random +0.001 (adversarial $\approx$30–40$\times$ larger when random is non-zero); Adult α=0.3 +0.38 vs +0.02 ($\approx$12–40$\times$). On LSAC the DP attack is weak in absolute terms because LSAC's clean DP is small (see §??).

k-NN graph ablation (Adult). The k-NN graph construction is available from `results/knn_ablation_k{5,10,15}.json`; the attack graph is insensitive to $k \in \{5, 10, 15\}$ (DP values across k are within $\pm$0.003 of each other at every α), so k=5 is a safe default for the IF attack.

Empirical radii (Q5). The bias-corrected clean-proportion formula $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1-2\alpha)$ is an empirical estimate, not a theoretical guarantee. Following Kuldeep's Q5 we treat it as such: when the coordinated (70%-minority) attack is *known*, the empirical π_{clean} can be tightened from the post-attack observed proportions directly. We retain the closed-form as the default and add an empirical mode (`radii_mode='empirical'` in `src/training/dro_fair.py::_compute_radii`) for the known-attack setting. This does *not* claim the paper is wrong. Full derivation in report appendix (Q5 math from B).

LSAC framing (Q3). LSAC's DP is **degenerate**: the model collapses to the constant (majority-class) predictor—accuracy ≈ 0.902, i.e. the 0.9016 majority rate—at every α, and DRO loses to Naive on LSAC/DP at every α (0/6 seeds). This is a *negative* result that we report openly, not a hidden failure of DRO: it is a consequence of LSAC's near-zero clean DP, which makes the DP-targeted attack ineffective. By contrast, LSAC/Combined is a genuine clean win (Wilcoxon p=0.0156 at α=0.1, 0.3, 0.4). The canonical τ=1, k_{inner}=10, 6-seed grid is complete (**540 rows**: DP+IF+Combined); Adult and Credit remain the defensible lead, with LSAC reported as split (Combined win / DP degeneracy / IF-attack DP loss).

Q7: IF $\leftrightarrow$ DP coupling (complete grid). The IF-attack third is complete (180 rows; cosine IF metric non-degenerate, max $|\text{if\_clean}| \approx 0.24$; full Wilcoxon in `results/if_wilcoxon_summary.txt`). Under the IF *attack*, DRO still lowers **DP** vs Naive on Adult at $\alpha \leq 0.2$ (6/6, p=0.0156) and on most Credit cells (Credit α=0.1 is 4/6, n.s.). On LSAC under IF attack DRO *loses* on DP (0/6 at $\alpha \leq 0.3$). At Adult $\alpha \geq 0.3$ the DP win under IF attack breaks even when the IF metric itself improves — the two criteria are coupled but not interchangeable. We do *not* claim a clean three-attack sweep on every dataset.

Statistical significance. The canonical grid uses n=6 seeds (0–5). Wilcoxon signed-rank one-sided p<0.05 holds at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks (verified in `results/canonical_tau1.json`). The full attack grid is complete (540 rows). Under IF attack the DP test is significant on Adult $\alpha \leq 0.2$ and most Credit cells; LSAC/IF and Adult IF $\alpha \geq 0.3$ do not support a DP win (see summary file).

4.2 Image data (UTKFace) — real features local; full grid pending

UTKFace is the third task of the project. Real ResNet-style features are now extracted locally (`data/raw/utkface_features.npz`); a multi-seed Fairness-Targeted PGD grid is running post-canonical-clear and is **not** claimed in this draft. Prior synthetic-only smoke outputs remain non-results; the older "DRO inverts on image features" claim stays *withdrawn.* flair2 GPU access is still optional once SSH is available.

5 Discussion

5.1 The Temperature (τ) Effect

Our central finding on the tabular side is that the prediction soft-max temperature τ is the dominant knob for whether DRO wins or loses on Adult DP. Under a stepped schedule (τ=100 for $\alpha \leq 0.3$, τ=1 for α=0.4) DRO loses on Adult DP almost everywhere; under a fixed τ=1 schedule DRO wins on Adult DP at $\alpha \leq 0.2$ (no method claim is made at $\alpha \geq 0.3$, where both methods fall below the constant predictor). At $\alpha \leq 0.2$ the gap widens monotonically. This is consistent with the theory: τ=100 sharpens the predictions toward 0/1 and so makes the inner re-weighting concentrate weights on a few “worst” samples, driving λ_{DP} to its clamp and causing the λ-runaway the earlier “adversarial feedback loop” discussion was observing. At τ=1 the soft predictions distribute weight smoothly and λ stays bounded, so the re-weighting helps rather than hurts.

5.2 Adversarial vs. Random Corruption

The empirical comparison (adversarial-vs-random runs) confirms that adversarial PGD raises DP 12–40× more than random label noise at matched α on Adult, and 2–22× on Credit. This rules out the “attack is too weak” counter-explanation for the Adult low-α cells.

5.3 Mechanism (old “feedback loop”)

The earlier “adversarial feedback loop” account of Adult $\alpha \geq 0.3$ (DRO collapsing to constant predictions) is no longer the right diagnosis under τ=1. We do still see lower accuracy at α=0.4 (≈ 0.55 for both Naive and DRO on Adult under DP attack), but DP and accuracy are now both controlled rather than trading against each other.

5.4 Empirical Radii (Q5)

The TV-radius formula $\rho_{\text{DP},j} = \alpha/((1-\alpha)\pi_j + \alpha)$ and the bias-corrected $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1-2\alpha)$ are empirical, not theoretical. Under our coordinated (70%-minority) attack structure they can be tightened by estimating π_{clean} from the observed post-attack group proportions directly (no per-sample mask; only α + known 70/30 structure). We implement this as `radii_mode='empirical'` in `src/training/dro_fair.py` (default stays 'uniform'). This is in line with Kuldeep's Q5 ("if the attack is known then we can use this approximation according to the attack") and does not claim the paper is wrong. See report appendix for the clean inversion derivation: $\pi_{\text{clean}}[\text{min}] = \pi_{\text{obs}}[\text{min}] + 0.4\alpha$, etc. (B's math).

5.5 Image data (UTKFace) — future work, no real run

The earlier "DRO inverts on ResNet18 features" claim from prior versions of this paper was based on tau=100-stepped results and is withdrawn; the only UTKFace outputs to date are synthetic smoke tests (random Gaussian features), since GPU access was never granted. The modality question remains open. See §4.2.

6 Related Work

Solans et al. [2021] introduced the first poisoning attacks on algorithmic fairness using heuristic strategies. Our work extends this with exact gradient computation and PGD optimization. Hashimoto et al. [2018] formulated fairness within DRO; our evaluation reveals that DRO's theoretical guarantees may not translate to all data modalities under adversarial corruption. Madry et al. [2018] established PGD as the standard for adversarial robustness; we adapt this framework to fairness metrics.

7 Conclusion

We introduced FairnessTargetedPGD, the first gradient-based adversarial attack targeting fairness metrics. On Adult and Credit, fixing $\tau = 1$ across all corruption levels makes DRO-FAIR achieve significantly lower DP than Naive-FAIR at $\alpha \leq 0.2$ under the DP-targeted and Combined attacks (Wilcoxon $p < 0.05$, $n = 6$ seeds; Adult/DP α=0.1 is 5/6), with accuracy equal-or-better. The earlier "DRO is fragile / inverts on Adult" finding was a high-temperature artifact (the stepped τ=100 schedule for $\alpha \leq 0.3$); switching to fixed τ=1 inverts that conclusion. LSAC/DP is a degenerate negative result (the model collapses to the constant predictor at every α, and DRO loses to Naive at 0/6 seeds); LSAC/Combined is a genuine win (p=0.0156). The full IF-attack third is complete (180/180; cosine IF non-degenerate): under IF attack DRO still wins on DP for Adult at $\alpha \leq 0.2$ and most Credit cells, but *loses* on LSAC/IF DP and at Adult $\alpha \geq 0.3$ the DP story breaks — not a blanket three-attack sweep. Adversarial PGD raises DP 12–40× more than random label noise on Adult, and the k-NN graph choice does not affect the DP attack. UTKFace real features are available locally; a full multi-seed image grid is in progress and is not claimed here. The earlier synthetic-only "DRO inverts on image features" claim remains withdrawn. Practical takeaway: when deploying DRO-FAIR, fix the prediction temperature and re-validate rather than relying on a τ-schedule.

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